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Music Genre Trend Prediction Based on Spatial-Temporal Music Influence and Euclidean Similarity

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Abstract—This paper is concerned with the music genre trends prediction according to the music influence and similarity through data analysis. By a Deepwalk and Cosine Similarity method, a mathematical model is formulated for the music influence, which evaluates the influence of musicians and between musicians. The music similarity model is proposed based on the Principal Component Analysis and Euclidean distance, which summarizes some phenomena according to genres. The genre-independent influence reflects the information about cultural influence. The visual demonstration of influence and similarity are provided. Finally, the trends of music genre are summarized and predicted from the music influence and similarity.

Index Terms—spatial influence, temporal influence, music similarity, music genre trends

I. Introduction

Music is an essential component of cultural heritage and has developed dozens of diverse genres. As a part of the world cultural heritage and human civilization, music can not only affect human behavior, but also affect human emotions[4]. Moreover, humans have a close relationship with music. In fact, the search of the world's modern music history reveals that the world of music has changed dramatically due to a series of social changes and the concerted efforts of influential artists[9].

Music and society influence each other over time. Music affects social development indirectly[5]. For humans, rhythm and melody can reach the depths of the soul and touch the heart, which affects the development of individuals, and people are the basic components of society[8]. Therefore, music can indirectly affect social development by influencing human intelligence.

The style of music is influenced by many factors. There are many factors that can influence musicians when they create a new piece of music, including their innate ingenuity, current social or political events, access to new instruments or tools, or other personal experiences[6]. Some people suggest that the influence of musicians can be measured by the characteristics of their music[7]. But the characteristics are difficult to recognize. Therefore, the measurement of music needs a criterion.

In this paper, the evaluation indicators are established firstly, which is composed of influence and similarity models. Influence includes spatial influence and temporal influence. Spatial influence indicates the breadth of a musician's influence and temporal influence reflects the depth of a musician's influence. There are a large number of unnormalized features and many existing similarity indicators, so it is necessary to perform principal component analysis and reduce the dimensionality of features, and propose a calculation method of music similarity. Then we conduct data analysis from the perspectives of influence and similarity, and explore the connection between observations in real phenomena.

Our goal is to predict the evolution and revolutionary trends of music genres by proposed two novel models: Spatial-Temporal Music Influence Model (STMI) and Euclidean Similarity Model (ESM). The STMI is inspired by Deepwalk[1] and Cosine Similarity, which evaluates the influence of musicians and among musicians. The similarity model ESM is proposed based on the Principal Component Analysis and Euclidean distance, which obtains some phenomena according to genres.

II. Data Analysis Model

A. Spatial-Temporal Music Influence

In the development of music, there is a phenomenon that artists always follow the great pioneers and influence the creation of the later generation. On the other hand, an artist's musical influence is not limited to the genre, but can also influence other genres. What is more, the influence may be large over time. So the model is established from two aspects: spatial influence and temporal influence. In order to look for spatial structure and temporal structure, a novel model named Spatial-Temporal Music Influence (STMI) is proposed, and the framework formula for the influence of the two musicians' models is as follows:

$$I_{Genre}^{i\leftarrow j} = \alpha I_S^{i\leftarrow j} + \beta I_T^{i\leftarrow j} \tag{1}$$

$$I_{Cross-genre}^{i\leftarrow j} = \gamma I_S^{i\leftarrow j} + \theta I_T^{i\leftarrow j} \tag{2}$$

Where I_S is spatial influence; I_T is temporal influence.

- Spatial Influence.

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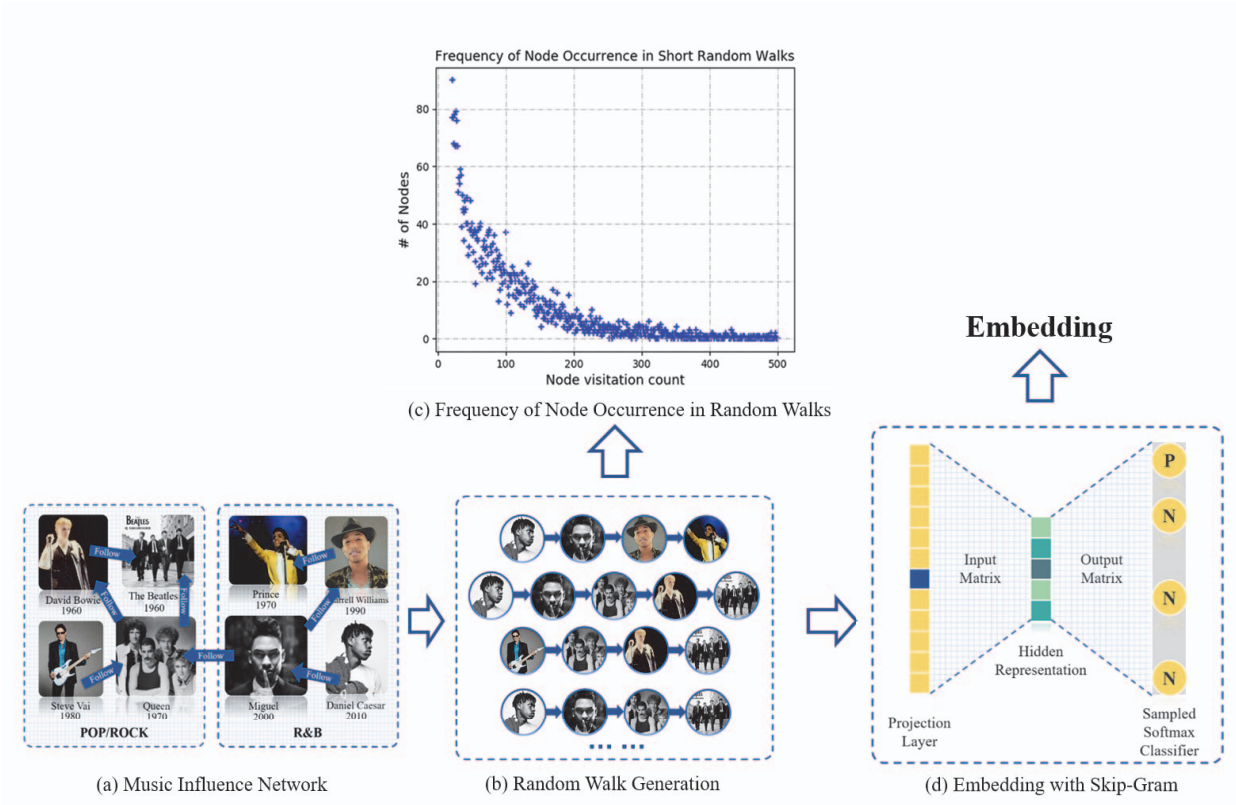

Fig. 1. Graph Embedding in The Music Influence Network

In order to obtain a specific influence value, the method of graph embedding is adopted to extract the structural features of the music influence network. As shown in Figure 1(c), the music influence network also conforms to the power law [2], that is, there are fewer heavyweight musicians and more ordinary musicians. As a result, the Deepwalk method is used to capture the features of each node. The embedding of nodes (i.e. musicians) are obtained by Deepwalk containing the structural characteristics of the music influence network. This process is called graph embedding, and its workflow is shown in Figure 1. Specifically, it is divided into two steps: (i)Random walk: random walk N times in the data from survey to get a sequence of length l, such as Figure 1(b). (ii) Skip-gram [3]: a neural network based way to convert words into feature embeddings. In this method, the sequence is treated as a sentence and the node is treated as a word.

In real life, the followers are often similar to the influencers. In general, cosine similarity can represent the correlation of two musicians i and j, which is formulated as

$$\cos(i,j) = \frac{\langle v_i, v_j \rangle}{|v_i||v_j|} \tag{3}$$

where $\langle \cdot, \cdot \rangle$ is the inner product operation of two s; v_i and v_j represent the feature embedding of the musicians respectively.

However, the influence is usually direct, and the influence of famous and ordinary musicians are different. Therefore, improvements are made based on Cosine Similarity. We assume that ordinary musicians are more easily influenced by famous musicians. On the contrary, the influence of ordinary musicians is relatively limited when famous musicians follow ordinary musicians. Simply put, in-degree is used to indicate the importance of musicians. Hence, taking musician j following musician i as an example, the following spatial influence model is proposed:

$$I_S^{i \leftarrow j} = \left| \frac{\langle v_i, v_j \rangle}{1 + \frac{D_j}{D_i}} \right| \tag{4}$$

Where v_i is the feature vector of node i and D_i is the node degree. When $\frac{D_j}{D_i}$ is larger, the spatial influence of i on j is larger.

- Temporal Influence.
  In a long time, music has changed with the development of society and the improvement of people's cognition. So musicians from different eras have less interaction with each other, and their influence decreases overtime. We propose the following temporal influence formula:

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$$I_T^{i \leftarrow j} = 1 - \frac{|\Delta t(i,j)|}{\sum_{k \in \mathbf{N}_i} |\Delta t(i,k)|} \quad (5)$$

where $\mathbf{N}_i$ represents the set of adjacent nodes of i. Considering space and time factors, we combine equations 1, 4, and 5 to propose STMI

$$I_{ST}^{i \leftarrow j} = \alpha(|\frac{v_i v_j}{1 + \frac{D_j}{D_i}}|) + \beta(1 - \frac{|\Delta t(i,j)|}{\sum_{k \in \mathbf{N}_i} |\Delta t(i,k)|}) \quad (6)$$

Further, the influence of a music artist is calculated by the following equation:

$$I_{ST}^{i} = \sum_{k \in \mathbf{N}_i} I_{ST}^{i \leftarrow k} \quad (7)$$

and it can be specifically divided into influences within and between genres.

B. Euclidean Music Similarity Model

Since ancient times, genres of music have been classified in many different ways. But the classifications are often controversial. Some similar styles are usually overlapped. There is no doubt that similarities can be found in many pieces of music, but musical similarities are very vague. Therefore, the measurement of musical similar needs a criterion.

- Dimensionality Reduction.
  Each kind of music has its own feature. There are 13 musical indexes that include energy, valence, tempo, etc. But the units between the indexes are not uniform and it is difficult to normalize directly. Obviously, using it directly is not wise. As a result, Principal Component Analysis (PCA) is used to retain the most important features of high-dimensional data and remove some unimportant features. The main idea of PCA is to map the n-dimensional features to the k-dimensional features, which are reconstructed on the basis of the original n-dimensional features. The python library3 is used to implement PCA algorithms.
  Features are gone down to 12 dimensions. The results show that after dimensionality reduction only the first eigenvalue is the most influential eigenvalues. This shows that reducing the dimension to 1 can express enough information. But the high dimension of musical features is reduced to 2 dimensions so as to facilitate the explanation of the subsequent work. Musical features as the basis for PCA dimension reductio will retain the necessary parameters in dimension reduction. At the same time, these parameters are used to reduce the dimension of the features of musicians.
- Establishment of Music Similarity.
  In general, Cosine Similarity, Pearson Correlation can be selected to measure the cosine similarity.

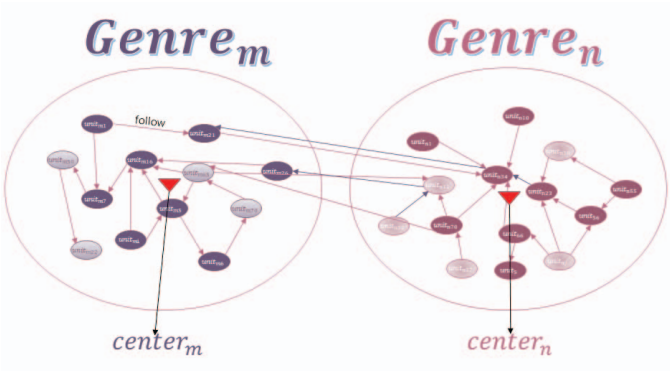

Fig. 2. Genres {m,n} and Central Points

However, there are only two dimensions in our whole dimensions, and what is really valid is just one dimension. When the differences are concentrated in one dimension, using cosine similarity is not suitable anymore. But the Euclidean distance means that the smaller the value is the more similar is. This law is inconsistent with our conventional intuition. Therefore, we improve the Euclidean distance by proposing a calculation method of music similarity called Euclidean Music Similarity(EMS).

$$f(u_1, u_2) = 1 - \frac{\sqrt{(u_{1,1} - u_{2,1})^2 + (u_{1,2} - u_{2,1})^2}}{max(\sqrt{(u_{i,1} - u_{j,1})^2 + (u_{i,2} - u_{j,1})^2})} \quad (8)$$

where $\{u_1, u_2\}$ are the artist units or center units,unit i and j represent any two artists.

- The Genre-Dependent and Genre-Independent Similarity.
  Through the previous method, the features of music and musicians, and the similarities among them have been determined. Next, a method is proposed to get the Genre-Dependent and Genre-Independent similarity is analyzed. The skeleton of our method is shown in Figure 2. Suppose Genrem=unit1,unit2...unitm, where M and N represent the number of artists in a genre, unit represents the features of a musician, center represents the features of the center.
  To be specific, a central point is chosen as the feature of genre. By calculating the average of all the songs' musical indexes that musicians sends is the best way to express the features of genre. Taking Genrem as an example, and the formula for calculating the center of feature is as follows:

$$C_m = \frac{1}{M} \sum_{i=1}^{M} unit_i^m \quad (9)$$

Similarly, Cn is figured out. Therefore, the genre-dependent similarity is calculated by the following

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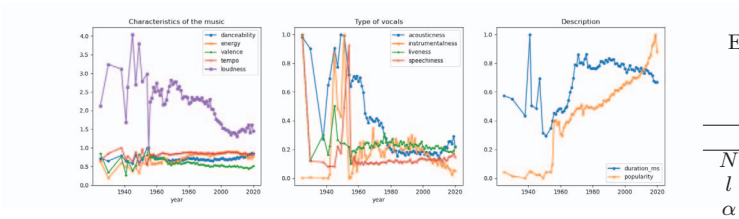

Fig. 3. Variation of Pop Rock

equation:

$$S_m = \frac{1}{M}\sum_{i=1}^{M} f(C_m, unit_i^m) \tag{10}$$

On the other hand, genre-independent similarity EMS equation is:

$$S_{m,n} = \frac{1}{2}[\frac{1}{M}\sum_{i=1}^{M} f(C_n, unit_i^m) + \frac{1}{N}\sum_{j=1}^{N} f(C_m, unit_j^n)] \tag{11}$$

C. External Influence of Music Model

Musicians' works are closely related to their personal thinking, emotion and cognition, but they are also inseparable from the influence of the society. Obviously, music expresses information about culture in circumstances. External Influence of Music Model is built to express information about cultural influence of music.

The music year data is analyzed and the data is visualized. Taking Pop Rock as an example, as shown in Figure 3.

The influence of society on music is a continuous process, so it is easy to observe changes in figure. Observing the Figure 3, we can see that the turning point in the influence of Pop Rock music was around 1960. The popularity of Pop Rock music began to decline from 1960. A historical and cultural analysis shows that the influence is reflected on various aspects of American society, including the attitude toward visible minority, time spirit and values[10]. Rock and roll played a non-ignorable role in the change of American traditional values. This shows that our model can reflect the influence of social factors on musical influence. In other words, social changes result in changes in several musical indexes within a music genre, leading to sudden changes in the general influence of that music genre.

III. Visual Demonstration

A. Experiment of Music Influence

- Experimental Settings.
  The following table is specific parameters set by our experiment. Note that since we believe that the time factor is more important between genres, the weight value is higher.
- Indexes That Determine Changes in Music Genres.

TABLE I
Experimental Parameters of Spatial-Temporal Music Influence Model

	Data	Meaning
N	50	Random walk rounds
l	5	Random walk sequence length
α	1.0	Weight coefficient of spatial influence within genre
β	1.0	Weight coefficient of temporal influence within genre
θ	1.0	Weight coefficient of spatial influence between genre
γ	1.5	Weight coefficient of temporal influence between genre

we use music feature indexes(e.g., danceability, energy) and influences of different years. We build the influence-feature model as follows:

$$I(t) = w_1 \times m_1(t) + w_2 \times m_2(t) + ... + w_k \times m_k(t) \tag{12}$$

Explanation of the parameters:

- k is the number of music feature indexes.
- $\{m_1(t),\ m_2(t)...\ m_k(t)\}$ are the corresponding feature indexes of music in year t.
- $\{w_1, w_2, ..., w_k\}$ are the quantities to be estimated from the data.

Based on 12, we can substitute the data to calculate the weight coefficient. Obviously, when the weight coefficient is larger, it means that the corresponding music feature is more important for influence.
The influence of music genre is divided into genre-dependent influence and genre-independent influence. The calculation results are shown in Table II and III, respectively. We observe the change of important indexes, and then compare this change with the trend of the influence of genre. These two trends are consistent with the feature that corresponding to this index is closely related to the change of the influence of the genre, that is to say, this feature is very important to the genre. As a result, the feature can help us to predict the genre trend.

TABLE II
The Weight of Musical Indexes within genres

scope	w_1	w_2	w_3
feature	danceability	energy	valence
value	-0.72	-0.58	0.60
scope	w_4	w_5	w_6
feature	acousticness	instrumentalness	livenesss
value	0.85	-0.26	0.67

TABLE III
The Weight of Musical Indexes between genres

scope	w_1	w_2	w_3
feature	danceability	energy	valence
value	-0.49	-0.43	0.20
scope	w_4	w_5	w_6
feature	acousticness	instrumentalness	livenesss
value	0.32	0.04	0.49

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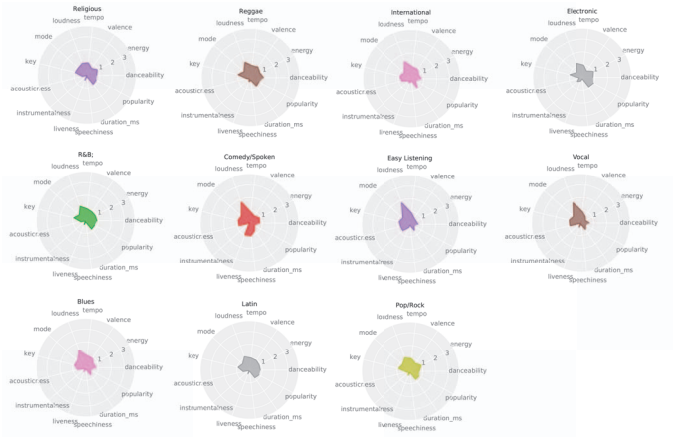

Fig. 4. Indexes of Different Genres

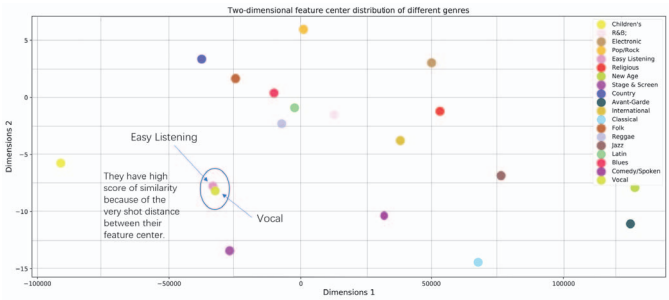

Fig. 5. Center Distribution

B. Experiment of Music Similarity

- Similarity Feature.
  Music features important indexes to distinguish music, so we distinguish music from the perspective of similarity. the original features (e.g. danceability) are visualized to calculate the similarity by category. Due to the inconsistent units of features, the data is normalized as shown in Figure 3.
- Genres' Feature Centers
  Figure 4 shows distribution of different genres' feature centers on 2 dimensions. We can observe that the feature distribution between genres is mainly reflected in only 1 dimension, it indicates that enough information has been retained by using PCA to reduce the indexes of music to 2 dimensions.
  Figure 6 shows the matrix of similarity within and among the genres. This information is visualized in the form of heat maps, containing information from two aspects. Colors on the main diagonal of the matrix show the genre-dependent similarity, and the other colors show the genre-independent similarity.

IV. Trends in Genre Change

Music genres with the development of society and the change of human cognition. In the process of music development and change, the influence and similarity of genres also change, we analyze separately below.

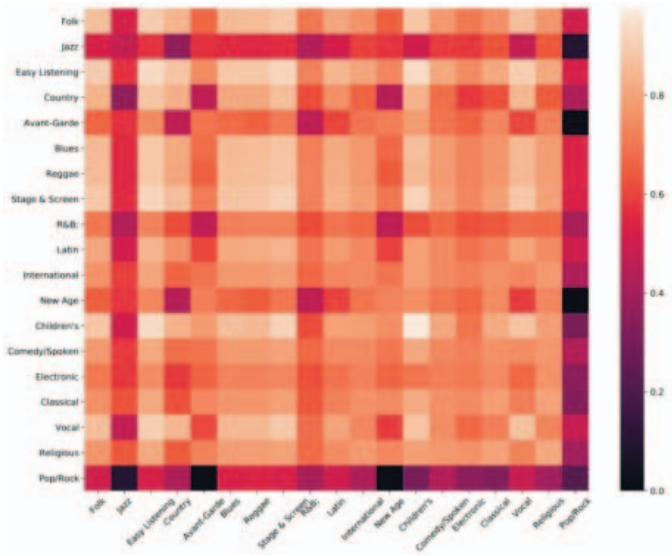

Fig. 6. Similarity Matrix: the value on the diagonal is the genre-dependent similarity, and the rest is the genre-independent similarity.

A. Change In Influence.

Music influence includes the genre-dependent influence and the genre-independent influence. So we analyze the change of influence over time from these two aspects:

- The genre-dependent influence grows over time, it means that the musical features of the genre become more similar. The walls of their circle grew thicker.
- The genre-independent influence increases as time goes, which means that the popularity of this genre increased gradually.

B. Change In Similarity.

Music similarity includes the genre-dependent similarity and genre-independent similarity. From these two aspects, Figure 7 shows how similarity changes over decades. Several representative genres are chosen to show. Changes are extracted from every 10-year heat map of the similarity matrix. Colors represent the score of similarity. The lighter the color, the higher the similarity.

V. Conclusion

Music influence and similarity are two indications to express music genre. In this paper, the Spatial-Temporal Music Influence Model is proposed to evaluate the influence from space and time. The Euclidean Similarity Model is constructed to measure music similarity. Then the data analysis showed that using music influence and similarity to predict music trends. The important influence feature and similarity indicators can predict the genre trend. The more genre-dependent influence, the music genre become more similar. The genre-independent influence increases means that the popularity of this genre increased. After combining the social environment and our model, it is concluded that the important index of genre changes consistent with the trend of the general index of music.

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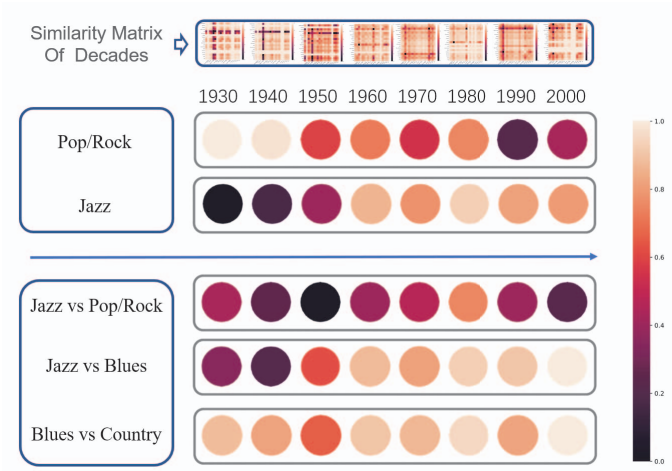

Fig. 7. Similarity Over Decades

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